Survival to 21s — Dev Session Summary

Project: KenPlay.Fun | Date: 2026-03-06 | Repo: github.com/zhumingh/survival-to-21s | Generated by Claude Code

AT A GLANCE

7	4	~500	67
COMMITS PUSHED	FILES CHANGED	LINES ADDED	COUNTRIES ADDED

COMMITS — TODAY'S WORK

Optimize game loop and fix core gameplay issues `c694cb5`

- **PERF** Replaced `setInterval(50ms)` with `requestAnimationFrame` — smooth 60fps rendering
- **PERF** Eliminated all `getBoundingClientRect()` in hot path — enemy x/y/size stored as numbers, zero layout thrashing
- **PERF** Delta-time applied to all movement — frame-rate independent speed
- **FIX** Helper/decoy feature was broken — attracted enemies now actually chase the decoy (stored helperPos as {x,y})
- **UX** Touch controls: replaced 8-directional swipe detection with smooth follow-finger movement
- **UX** Score display changed from `5s042` to `5.04s`
- **UX** Milestone flash alerts at 10s / 20s / 30s speed-up events

Overhaul game-over screen with vivid layered stats `ec45e2b`

- **UX** Animated time counter counts up from 0 to final score over 1.3s
- **UX** Star rating (1–5 ★) with staggered spring-pop animation based on survival time
- **UX** Rank label: JUST STARTED → KEEP GOING → NOT BAD → IMPRESSIVE → LEGENDARY (≥21s)
- **UX** 2×2 stat grid: steps taken, enemies dodged, closest call (px), proximity label
- **UX** Milestone badges (gold pills) for 10s / 20s / 30s actually survived
- **PERF** Track `closestCall` (min enemy distance) and `enemiesDodged` per run — no overhead

Port all improvements to the live server (public/) `21e58d0`

- **PERF** All performance fixes applied to `public/js/game.js` (live server) — not just the standalone prototype
- **FIX** Helper/decoy fix ported — glowing box-shadow added to helper element

- UX Follow-finger touch controls ported to live game
- UX Full game-over panel with CSS animations added to `public/css/style.css`
- UX `#milestone` div added to `public/index.html` for in-game speed-up flash

Fix milestone flash duration and game-over screenshot safety

`ed7f0db`

- FIX SPEED UP / FASTER / MAXIMUM SPEED flash shortened from 1800ms → 700ms
- FIX Restart now requires **Enter or Space** (not any key) — screenshot shortcuts (Cmd+Shift+3/4/5) no longer accidentally trigger next round

Show scaled player route trace inside game-over panel

`f699d0b`

- FEAT New `drawPathInPanel()` — fits the full movement path into a 370×200 canvas in the game-over screen
- FEAT Path auto-scaled and centered regardless of how much area was explored
- FEAT Subtle grid background, green glow on path, green dot = start, red dot = death point
- UX Labeled "Your route this round" — designed for screenshotting and sharing

Add country profiles and country leaderboard

`7db626c`

- DB Safe migration: `ALTER TABLE users ADD COLUMN country TEXT` — works on existing databases
- DB Register endpoint validates and stores ISO-2 country code
- DB New API endpoint: `GET /api/leaderboard/countries` — best score, avg score, player count per country
- FEAT Country dropdown on registration (67 countries with flag emojis), hidden during login
- FEAT Global leaderboard: flag emoji + country name column per player
- FEAT New Countries tab in leaderboard — nations ranked by best player score
- UX In-game HUD shows flag emoji next to logged-in username
- UX Player's own country row highlighted in green in Countries tab

Show dedicated Country column in global leaderboard tab

`622f5bd`

- UX Moved country from embedded in player name to its own column: flag + full country name
- UX Players without a country set show 🏳 —

FILES MODIFIED

File	What Changed
`ken_survival.html`	Standalone prototype — full game loop rewrite (rAF, stored positions, helper fix, touch controls, game-over panel, route trace)

File	What Changed
`public/js/game.js`	Live server game logic — all improvements ported + country data (67 entries), countryFlag(), leaderboard rendering with flags, Countries tab
`public/css/style.css`	Game-over CSS animations (goTitlePulse, starPop, goSlideUp, restartBlink), country select styling, milestone flash styles
`public/index.html`	Added #milestone div, country select input group, Countries tab button in leaderboard
`server.js`	DB migration for country column, register/login/me endpoints updated, new /api/leaderboard/countries endpoint

GAME-OVER PANEL — LAYER STRUCTURE

Layer	Content	Animation
1	**GAME OVER** title	Infinite red glow pulse
2	Time counter	Counts up 0→final over 1.3s (rAF)
3	Route trace canvas	Static — fitted path with glow
4	Star rating (1–5)	Each star pops in with spring delay
5	Rank label	Slide up, color-coded by tier
6	Stats grid (6 cells)	Staggered slide-up
7	Milestone badges	Slide-up, only if earned
8	Restart prompt	Blinks after 2.5s — Enter/Space only

COUNTRY FEATURE — API & UI SUMMARY

Component	Detail
DB migration	`ALTER TABLE users ADD COLUMN country TEXT NOT NULL DEFAULT ''` — wrapped in try/catch, safe for existing DBs
Validation	Server validates country is exactly 2 uppercase letters (ISO 3166-1 alpha-2); invalid input stored as empty string
Country list	67 countries in `COUNTRIES` array, rendered as flag emoji + name in dropdown
Flag rendering	`countryFlag(code)` uses Unicode regional indicator symbols — no image assets needed

Component	Detail
GET /api/leaderboard/countries	Returns country, player count, best_ms, avg_ms — grouped by country, ordered by best score
Global tab	New Country column: flag + country name; 🏳️ — for users without country
Countries tab	# | Country | Players | Best | Avg — user's country row highlighted green

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